آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 4

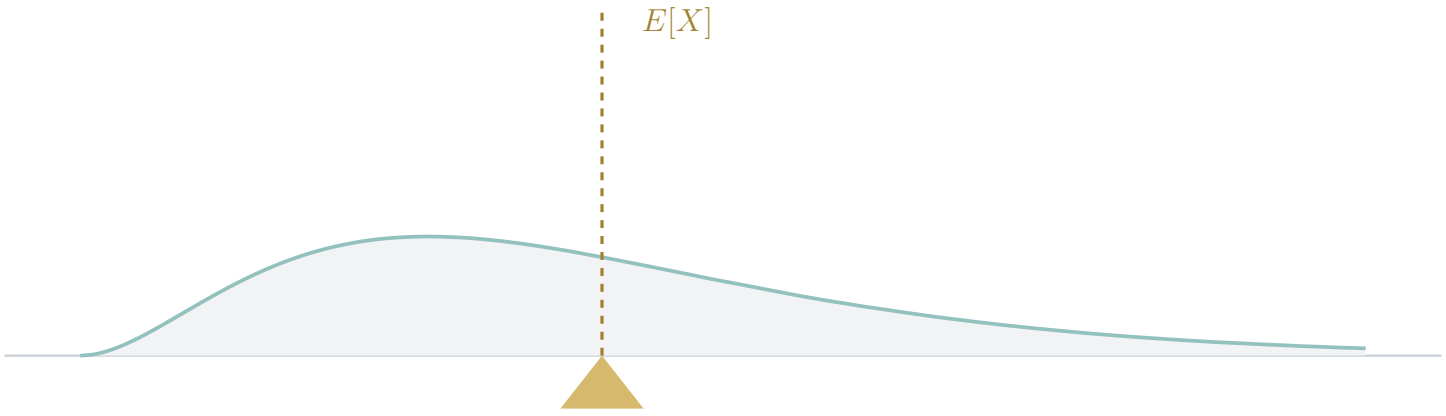
Expectation and Moments

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In This Module

4.1 Introduction: Why Summary Statistics? . . . 3
4.2 Expected Value (Mean) . . . 4
4.3 Expected Value of a Function: $E[g(X)]$ . . . 5
4.4 Variance . . . 6
4.5 Standard Deviation . . . 7
4.6 Properties of Expectation . . . 8
4.7 Properties of Variance . . . 9
4.8 Higher-Order Moments . . . 10
4.9 Moment Generating Function (MGF) . . . 11
4.10 Engineering Applications of Moments . . . 13
4.11 Practice Problems . . . 14
4.12 Key Takeaways . . . 17

Learning Objectives

After completing this module, students will be able to:

1. Compute expected values for discrete and continuous random variables
2. Calculate variance and standard deviation and interpret their engineering meaning
3. Apply properties of expectation and variance in engineering analysis
4. Compute and interpret higher-order moments
5. Use the moment generating function (MGF)
6. Relate statistical moments to physical engineering quantities

4.1 Introduction: Why Summary Statistics?

The Engineering Need

A probability distribution contains ALL information about a random variable. But for engineering decisions, we often need concise summaries:

- **Signal processing:** What is the average power? $\rightarrow$ Mean of X^2
- **Quality control:** How much *do* components vary? $\rightarrow$ Variance
- **Communication:** What is the noise power? $\rightarrow E[N^2]$
- **Reliability:** What is the average lifetime? $\rightarrow$ Mean

Moments (mean, variance, etc.) are the essential **engineering-relevant summaries** of a distribution.

4.2 Expected Value (Mean)

Definition — Discrete RV

$$E[X] = \mu_X = \sum_x x \cdot p_X(x)$$

Sum of (each value $\times$ its probability).

Definition — Continuous RV

$$E[X] = \mu_X = \int_{-\infty}^{\infty} x \cdot f_X(x)\, dx$$

Engineering Interpretation

The expected value is the **long-run average**: if you repeat the experiment many times and average the results, you get approximately $E[X]$.

Engineering Quantity	Expected Value Meaning
Noise voltage $E[N]$	DC bias/offset in noise
Signal amplitude $E[A]$	Average signal level
Component lifetime $E[T]$	Mean time to failure (MTTF)
Packet delay $E[D]$	Average delay
Bit errors $E[X]$	Average error count

Example: Average Noise Power

For zero-mean Gaussian noise $N \sim N(0, \sigma^2)$:

- $E[N] = 0$ (no DC component)
- Average power $= E[N^2] = \sigma^2$ (this is the variance!)

Example: Expected Number of Retransmissions

ARQ protocol: each transmission succeeds with probability $p = 0.8$.
Number of attempts $X \sim$ Geometric(0.8).
$E[X] = \frac{1}{p} = \frac{1}{0.8} = 1.25$ attempts on average.

4.3 Expected Value of a Function: $E[g(X)]$

The LOTUS Theorem (Law of the Unconscious Statistician)

To find $E[g(X)]$, you *do* NOT need the distribution of $Y = g(X)$:

Discrete:

$$E[g(X)] = \sum_x g(x) \cdot p_X(x)$$

Continuous:

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx$$

Engineering Applications

Function $g(X)$	Engineering Meaning
$g(X) = X^2$	Mean square value (power)
$g(X) = (X - \mu)^2$	Variance (AC power)
$g(X) =$	X
$g(X) = e^{j\omega X}$	Characteristic function

Example: Mean Power of a Signal

Signal X has PDF $f_X(x)$. Power $P = X^2$.
$E[P] = E[X^2] = \int x^2 f_X(x)\, dx$
For $X \sim N(0, \sigma^2)$: $E[X^2] = \sigma^2$ (the noise power equals the variance).
For $X \sim N(\mu, \sigma^2)$: $E[X^2] = \mu^2 + \sigma^2$ (total power = DC power + AC power).

4.4 Variance

Definition

$$\text{Var}(X) = E[(X - \mu_X)^2] = E[X^2] - (E[X])^2$$

The second formula (computational formula) is often easier to use.

Derivation of Computational Formula

$$\begin{aligned}
&\text{Var}(X) = E[(X - \mu)^2] \\
&= E[X^2 - 2\mu X + \mu^2] \\
&= E[X^2] - 2\mu E[X] + \mu^2 \\
&= E[X^2] - 2\mu^2 + \mu^2 \\
&= E[X^2] - \mu^2
\end{aligned}$$

Engineering Interpretation

The variance measures **spread/dispersion/uncertainty**:

Context	Variance Meaning
Noise signal	Noise power (for zero-mean noise)
Manufacturing	How much components deviate from nominal
Measurements	Measurement precision (smaller = more precise)
Signal	AC power component
Lifetime	How predictable is the lifetime?

Key Insight: Variance = Power in Engineering

For a zero-mean signal X:

- $E[X^2] = \text{Var}(X) = \sigma^2$ = **total power**
- For $X \sim N(0, \sigma^2)$: noise power $= \sigma^2 = \text{Var}(X)$

For a signal with DC offset ($\mu \neq 0$):

- $E[X^2] = \mu^2 + \sigma^2$ = **total power = DC power + AC power**
- $\text{Var}(X) = \sigma^2$ = **AC power only**

4.5 Standard Deviation

Definition

$$\sigma_X = \sqrt{\mathrm{Var}(X)}$$

Engineering Significance

The standard deviation has the **same units** as X, making it directly interpretable:

Random Variable	Standard Deviation Unit	Interpretation
Voltage noise	Volts	RMS noise voltage
Timing jitter	Seconds	RMS jitter
Distance error	Meters	RMS position error
Temperature	°C	RMS temperature fluctuation

The standard deviation is the **RMS (root-mean-square)** value for zero-mean signals.

4.6 Properties of Expectation

Linearity (FUNDAMENTAL)

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y] \quad \text{(ALWAYS, even if dependent!)}$$

$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i]$$

Monotonicity

If $X \geq 0$ always, then $E[X] \geq 0$.

If $X \geq Y$ always, then $E[X] \geq E[Y]$.

For Independent RVs

If X and Y are independent:

$$E[XY] = E[X] \cdot E[Y]$$

⚠ This does NOT hold for dependent variables!

Engineering Example: Total Noise

A received signal: $R = S + N_1 + N_2$ (signal plus two noise sources)
$E[R] = E[S] + E[N_1] + E[N_2] = E[S] + 0 + 0 = E[S]$
(Zero-mean noise doesn't bias the expected received signal.)

4.7 Properties of Variance

Scaling and Shift

$$\text{Var}(aX + b) = a^2\text{Var}(X)$$

- Adding a constant b does NOT change variance (shifting doesn't change spread)
- Scaling by a multiplies variance by a^2

Sum of Independent RVs

If $X_1, X_2, \ldots, X_n$ are **independent**:

$$\text{Var}(X_1 + X_2 + \cdots + X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_n)$$

⚠ Variances add for INDEPENDENT variables only!

Engineering Example: Noise Power Addition

Two independent noise sources ($\sigma_1^2 = 0.01\,\text{V}^2$, $\sigma_2^2 = 0.04\,\text{V}^2$):
Total noise power $= \sigma_1^2 + \sigma_2^2 = 0.05\,\text{V}^2$
Total RMS noise $= \sqrt{0.05} = 0.224\,\text{V}$
Note: RMS values *do* NOT add directly! $(0.1 + 0.2 \neq 0.224)$

Variance of Sample Mean

For n i.i.d. observations $X_1, \ldots, X_n$ with variance σ^2:

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n}\sum X_i\right) = \frac{\sigma^2}{n}$$

More measurements $\rightarrow$ lower uncertainty in the average.

4.8 Higher-Order Moments

$n-$ th Moment

$$\mu_n' = E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x)\, dx$$

$n-$ th Central Moment

$$\mu_n = E[(X-\mu)^n]$$

- $\mu_1 = 0$ (always)
- $\mu_2 = \text{Var}(X)$
- μ_3 relates to **skewness**
- μ_4 relates to **kurtosis**

Skewness ($3rd$ Central Moment, Normalized)

$$\gamma_1 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Skewness	Shape	Example
$\gamma_1 = 0$	Symmetric	Gaussian
$\gamma_1 > 0$	Right-skewed (long right tail)	Exponential
$\gamma_1 < 0$	Left-skewed (long left tail)	Negated exponential

Engineering significance: Skewness tells us if extreme values are more likely in one direction. Important for reliability (failure time distributions are right-skewed).

Kurtosis (4th Central Moment, Normalized)

$$\gamma_2 = \frac{E[(X-\mu)^4]}{\sigma^4} - 3$$

(Subtracting 3 makes Gaussian have kurtosis 0: "excess kurtosis.")

Kurtosis	Interpretation	Meaning
$\gamma_2 = 0$	Mesokurtic	Gaussian-like tails
$\gamma_2 > 0$	Leptokurtic	Heavier tails than Gaussian
$\gamma_2 < 0$	Platykurtic	Lighter tails than Gaussian

Engineering significance: High kurtosis means more outliers — important for impulsive noise modeling.

4.9 Moment Generating Function (MGF)

Definition

$$M_X(t) = E[e^{tX}]$$

Discrete: $M_X(t) = \sum e^{tx} p_X(x)$
Continuous: $M_X(t) = \int e^{tx} f_X(x)\, dx$

Why "Moment Generating"?

The $n-$ th moment is obtained by differentiating n times at $t = 0$:

$$E[X^n] = \left.\frac{d^n M_X(t)}{dt^n}\right|_{t=0}$$

- $M'_X(0) = E[X]$
- $M''_X(0) = E[X^2]$
- $\text{Var}(X) = M''_X(0) - [M'_X(0)]^2$

Properties

1. **Uniqueness:** If $M_X(t) = M_Y(t)$ for all t in a neighborhood of 0, then X and Y have the same distribution.
2. **Sum of Independent RVs:** If X and Y are independent: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$
3. **Linear transformation:** $M_{aX+b}(t) = e^{bt} \cdot M_X(at)$

MGFs of Common Distributions

Distribution	MGF $M_X(t)$
Bernoulli(p)	$1 - p + pe^t$
Binomial(n, p)	$(1 - p + pe^t)^n$
Poisson(λ)	$\exp(\lambda(e^t - 1))$
Exponential(λ)	$\frac{\lambda}{\lambda - t},\ t < \lambda$
Gaussian(μ, σ^2)	$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$

Engineering Application: Sum of Independent Signals

If signal components $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent:

$$M_{X_1+X_2}(t) = M_{X_1}(t) \cdot M_{X_2}(t)$$
$$= \exp\left(\mu_1 t + \frac{\sigma_1^2 t^2}{2}\right) \cdot \exp\left(\mu_2 t + \frac{\sigma_2^2 t^2}{2}\right)$$
$$= \exp\left((\mu_1 + \mu_2)t + \frac{(\sigma_1^2 + \sigma_2^2)t^2}{2}\right)$$

This is the MGF of $N(\mu_1 + \mu_2,\, \sigma_1^2 + \sigma_2^2)$. Therefore $X_1 + X_2 \sim N(\mu_1 + \mu_2,\, \sigma_1^2 + \sigma_2^2)$.

4.10 Engineering Applications of Moments

Application 1: Noise Power Characterization

A noise signal $N(t)$ is sampled. The samples have mean μ_N and variance σ_N^2.

- **DC component (bias):** μ_N — should be zero for ideal noise
- **AC noise power:** σ_N^2 — determines SNR
- **Total power:** $E[N^2] = \mu_N^2 + \sigma_N^2$

SNR (Signal-to-Noise Ratio):

$$\text{SNR} = \frac{E[S^2]}{\sigma_N^2} = \frac{\text{Signal Power}}{\text{Noise Power}}$$

Application 2: Measurement Error

A sensor measures true value θ with error E:

- Measurement: $X = \theta + E$
- $E[X] = \theta + E[E]$ — bias in measurement
- $\text{Var}(X) = \text{Var}(E)$ — measurement precision

Accuracy: $|E[E]|$ (how close the average is to truth)
Precision: σ_E (how repeatable measurements are)

Application 3: Component Lifetime Statistics

Components have lifetime $T \sim \text{Exponential}(\lambda)$:

- Mean Time To Failure: $\text{MTTF} = E[T] = \frac{1}{\lambda}$
- Standard Deviation: $\sigma_T = \frac{1}{\lambda} = \text{MTTF}$
- For exponential: coefficient of variation $\text{CV} = \frac{\sigma}{\mu} = 1$ (always!)

Application 4: Communication Signal Amplitude

Received signal $R = A + N$ where A = signal amplitude, $N \sim N(0, \sigma^2)$:

- $E[R] = A$ (signal amplitude preserved on average)
- $\text{Var}(R) = \sigma^2$ (noise power determines variance)
- $\text{SNR} = \dfrac{A^2}{\sigma^2}$

4.11 Practice Problems

Problem 1: Basic Computation

A discrete RV X has PMF: $P(X=-1)=0.2$, $P(X=0)=0.5$, $P(X=1)=0.2$, $P(X=2)=0.1$.

(a) Find $E[X]$.
(b) Find $E[X^2]$.
(c) Find $\text{Var}(X)$.
(d) Find $E[3X+2]$.
(e) Find $\text{Var}(3X+2)$.

Solution

(a) $E[X] = (-1)(0.2) + (0)(0.5) + (1)(0.2) + (2)(0.1) = -0.2 + 0 + 0.2 + 0.2 = 0.2$
(b) $E[X^2] = (1)(0.2) + (0)(0.5) + (1)(0.2) + (4)(0.1) = 0.2 + 0 + 0.2 + 0.4 = 0.8$
(c) $\text{Var}(X) = E[X^2] - (E[X])^2 = 0.8 - 0.04 = 0.76$
(d) $E[3X+2] = 3E[X] + 2 = 3(0.2) + 2 = 2.6$
(e) $\text{Var}(3X+2) = 9 \cdot \text{Var}(X) = 9(0.76) = 6.84$

Problem 2: Signal and Noise

A received signal $R = 2.5 + N$, where $N \sim N(0,\, 0.04)$.

(a) Find $E[R]$ and $\text{Var}(R)$.
(b) Find the SNR in dB.
(c) Find $P(R<0)$ — probability the signal goes negative.

Solution

(a) $E[R] = 2.5$, $\text{Var}(R) = \text{Var}(N) = 0.04$, $\sigma_R = 0.2$
(b) Signal power $= 2.5^2 = 6.25$, Noise power = 0.04
SNR $= \frac{6.25}{0.04} = 156.25 \rightarrow \text{SNR}_{dB} = 10 \cdot \log_{10}(156.25) = 21.94$ dB
(c) $P(R<0) = P(N<-2.5) = \Phi\left(-\frac{2.5}{0.2}\right) = \Phi(-12.5) \approx 0$ (extremely unlikely)

Problem 3: Averaging Measurements

A sensor makes n independent measurements of a constant voltage $V = 3.3\,\text{V}$.
Each measurement: $X_i = 3.3 + N_i$ where $N_i \sim N(0,\ 0.01)$.

(a) What is $E[\bar{X}]$ and $\text{Var}(\bar{X})$ for $n = 1$?
(b) How many measurements n needed so that $\sigma_{\bar{X}} < 0.01\,\text{V}$?
(c) With $n = 100$, what is $P(|\bar{X} - 3.3| > 0.02)$?

Solution

(a) $E[\bar{X}] = 3.3,\ \text{Var}(\bar{X}) = \dfrac{0.01}{1} = 0.01,\ \sigma = 0.1\,\text{V}$

(b) $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} < 0.01 \rightarrow \sqrt{n} > 10 \rightarrow n > 100 \rightarrow$ need $n \geq 101$

(c) $\sigma_{\bar{X}} = \dfrac{0.1}{\sqrt{100}} = 0.01\,\text{V}$
$P(|\bar{X} - 3.3| > 0.02) = P(|Z| > 2) = 2(1 - \Phi(2)) = 0.0455$

Problem 4: Component Lifetime

Components have lifetime $T \sim \text{Exp}(\lambda = 0.002$ per hour$)$.

(a) Find MTTF, σ_T, and coefficient of variation.
(b) Find $P(T > 1000\,\text{hours})$.
(c) Find $E[T^2]$ and interpret.

Solution

(a) MTTF $= \frac{1}{\lambda} = 500$ hours, $\sigma_T = 500$ hours, CV $= \frac{\sigma}{\mu} = 1$

(b) $P(T > 1000) = e^{-0.002 \times 1000} = e^{-2} = 0.1353$

(c) $E[T^2] = \text{Var}(T) + (E[T])^2 = 500^2 + 500^2 = 500000\,\text{hours}^2$
$\sqrt{E[T^2]} = 707$ hours (RMS lifetime)

Problem 5: MGF Application

X has MGF $M_X(t) = \frac{1}{3}e^t + \frac{2}{3}e^{2t}$.

(a) What type of RV is X? Find its PMF.

(b) Find $E[X]$ using the MGF.

(c) Find $\text{Var}(X)$ using the MGF.

Solution

(a) X is discrete. Comparing with $\sum p_X(x)e^{tx}$:

$$P(X=1) = \frac{1}{3},\ P(X=2) = \frac{2}{3}$$

(b) $M'(t) = \frac{1}{3}e^t + \frac{4}{3}e^{2t}$

$$E[X] = M'(0) = \frac{1}{3} + \frac{4}{3} = \frac{5}{3}$$

(c) $M''(t) = \frac{1}{3}e^t + \frac{8}{3}e^{2t}$

$$E[X^2] = M''(0) = \frac{1}{3} + \frac{8}{3} = 3$$

$$\text{Var}(X) = 3 - \frac{5}{3}^2 = 3 - \frac{25}{9} = \frac{2}{9}$$

4.12 Key Takeaways

1. $E[X]$ = **long-run average** — the center of the distribution, DC component
2. $\mathrm{Var}(X)$ = **spread/power** — for zero-mean signals, variance IS power
3. σ **has same units as** X — directly interpretable as RMS value
4. **Linearity of expectation** always holds; variance addition requires independence
5. **MGF uniquely identifies** a distribution and generates all moments
6. **In engineering:** mean $\rightarrow$ signal level, variance $\rightarrow$ noise power, std dev $\rightarrow$ RMS noise

Next Module: Module 5 — Important Random-Variable Functions